

HISABDO AI — DAY 18 WEB INTEGRATION

Professional MVP Specification, API Design & Implementation Guide

Source alignment: This Day 18 plan follows the established HisabDo architecture in the provided Day 10 specification: verified backend results are calculated outside the LLM, then reused by web/mobile UIs; the same backend result should be reusable by both applications. ■filecite■turn0file0■L183-L185■

Day 18 Objective

Integrate the verified HisabDo AI results into the web application and provide a complete, testable web experience for Financial Health, Spending Patterns, Anomalies, Cash-Flow Forecast, Budget Recommendations, AI Recommendations and Ask HisabDo AI.

1. Final Web Feature Scope

- AI Financial Dashboard
- Financial Health Score and status
- Spending Pattern Intelligence
- Anomaly Detection
- Cash-Flow Forecast
- Budget Recommendations
- AI Recommendations with priority/severity
- Recommendation detail view
- Ask HisabDo AI
- Loading, empty, error and retry states
- Responsive web layout

2. Architecture

HisabDo Database → .NET Calculation/Analytics Engine → Verified AI Results → Web API → Web UI. The LLM remains an explanation layer only. The supplied Day 10 specification explicitly states that financial totals, percentages, scores and forecast values originate from trusted HisabDo data and a deterministic calculation/analytics layer.

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3. Financial Health Web Card

Display: score, status, saving rate, expense ratio, budget status, cash flow, expense growth, and key insights. The Day 10 contract already defines these backend fields and says the same backend result should be reusable by Flutter mobile and web. ■filecite■turn0file0■L167-L180■ ■filecite■turn0file0■L183-L195■

4. Spending Pattern Section

Show category, current amount, previous amount where available, percentage change where verified, and direction. Never calculate a new financial percentage in the browser.

5. Anomaly Section

Show only verified anomalies from the backend: anomaly type, category, amount, severity and reason. If no verified anomaly exists, show an empty state rather than inventing a message.

6. Cash-Flow Forecast Section

Show forecast period, expected income, expected expenses, expected savings, expected closing balance and confidence. Forecast values come from the backend forecasting engine.

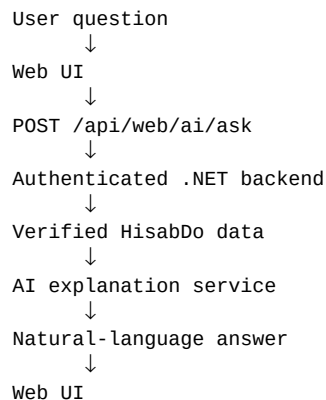
7. Budget Recommendation Section

Show category, recommended budget, existing budget when available, historical average and backend reason. The browser does not recalculate the recommendation.

8. AI Recommendation Section

Show recommendation summary, priority, severity where available, reason and action. Expected benefit is displayed only when the backend marks it as verified.

9. Ask HisabDo AI Web Flow



10. Web API

GET /api/web/ai/dashboard — returns the complete dashboard DTO. POST /api/web/ai/ask — accepts a natural-language question and returns an explanation based on verified data.

11. Loading / Empty / Error / Retry

- Loading: show skeleton/spinner and disable duplicate requests.
- Empty: explain that no verified data is currently available.
- Error: show a user-friendly error without exposing stack traces.
- Retry: repeat the failed request without refreshing the entire page.
- Partial data: render available sections and clearly mark unavailable sections.

12. Security and Data Isolation

The web backend must derive the user identity from authentication claims/session, retrieve only permitted records, and never trust a client-supplied user ID for authorization. This follows the established backend processing flow: authenticate user → retrieve only permitted records → calculate/aggregate → return verified API results.

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13. Team Coordination

Member	Day 18 focus
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Taha — Team Lead	Finalize requirements, review APIs, integration, acceptance and blockers
Jaffer	Web dashboard/UI implementation and API wiring
Omesha	Financial Health, Spending Pattern and data display validation
Aarti	AI explanation/recommendation presentation and Ask HisabDo flow
Laiba	Testing, error/empty states, responsive UI and acceptance evidence

14. Testing & Acceptance

- Dashboard loads with valid authenticated data.
- Financial Health score/status match backend response exactly.
- Spending pattern values match backend response exactly.
- Anomalies display only verified backend records.
- Forecast values match forecasting engine response.
- Budget recommendations match backend output.
- Recommendation priority/severity are not recalculated in the browser.
- Ask HisabDo answers use verified backend data.
- No AI-generated financial numbers are invented.
- No-data users receive safe empty states.
- API failures show safe error states and retry works.
- User A cannot access User B's financial data.
- Responsive layout works across desktop/tablet/mobile widths.

15. Implementation Files

The accompanying files are intentionally separate: a C# ASP.NET Core API controller/DTO contract, a JavaScript web integration client, and a standalone HTML dashboard prototype.

C# API: Day18_HisabDo_WebIntegration_DotNet.cs

JavaScript: Day18_HisabDo_WebIntegration.js

HTML prototype: Day18_HisabDo_WebDashboard.html

16. Important MVP Rule

The web application is a presentation/integration layer. Financial calculations remain in the trusted backend/calculation engine. The LLM explains verified results and does not determine financial numbers. This is consistent with the provided project specification's no-invention rule. ■filecite■turn0file0■L121-L139■